[BUG] qchem.electron_integrals IndexError: list index out of range

https://github.com/PennyLaneAI/pennylane/issues/3490

ROW 54

[BUG] qchem.electron_integrals IndexError: list index out of range #3490

New issue

Closed

yitchen-tim opened on Dec 7, 2022

Expected behavior

Receive the electron integrals of the molecule

Actual behavior

IndexError: list index out of range

Additional information

In qml.qchem module, there are many methods that return a function. For example, the `repulsion_tensor` and `electron_integrals`. The functions returned by these methods lack definition of how to use it. In this case, I don't know what should the input be for function returned by `electron_integrals`. I tried out different format for the inut, but I couldn't get it to work.

Source code

```
import pennylane as qml
import pennylane.numpy as np

symbols = ["H", "O", "H"]
geometry = np.array([
    [0.0399, -0.0038, 0.0],
    [1.5780,  0.8540, 0.0],
    [2.7909, -0.5159, 0.0]]
)
mol = qml.qchem.Molecule(symbols, geometry, basis_name="sto-3g")
ei = qml.qchem.electron_integrals(mol)(np.array(mol.alpha))
```

Tracebacks

```
----> 3 ei = qml.qchem.electron_integrals(mol)(np.array(mol.alpha))

~/anaconda3/envs/Braket/lib/python3.7/site-packages/pennylane/qchem/hamiltonian.py in _electron_integrals(*arg
    117             3,
    118         )
--> 119         core_constant = nuclear_energy(mol.nuclear_charges, mol.coordinates)(*args)
    120
    121         if core is None and active is None:

~/anaconda3/envs/Braket/lib/python3.7/site-packages/pennylane/qchem/hartree_fock.py in _nuclear_energy(*args)
    214         for i, r1 in enumerate(coor):
    215             for j, r2 in enumerate(coor[i + 1 :]):
--> 216                 e = e + (charges[i] * charges[i + j + 1] / anp.linalg.norm(r1 - r2))
    217         return e
    218

IndexError: list index out of range
```

System information

```
Name: PennyLane
Version: 0.27.0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /home/ec2-user/anaconda3/envs/Braket/lib/python3.7/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, retworkx, sc
Required-by: amazon-braket-algorithm-library, amazon-braket-pennylane-plugin, PennyLane-Lightning

Platform info:           Linux-5.10.102-99.473.amzn2.x86_64-x86_64-with-glibc2.17
Python version:          3.7.15
Numpy version:           1.21.6
Scipy version:           1.7.3
```

Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

Participants

```
Installed devices:
- default.gaussian (PennyLane-0.27.0)
- default.mixed (PennyLane-0.27.0)
- default.qubit (PennyLane-0.27.0)
- default.qubit.autograd (PennyLane-0.27.0)
- default.qubit.jax (PennyLane-0.27.0)
- default.qubit.tf (PennyLane-0.27.0)
- default.qubit.torch (PennyLane-0.27.0)
- default.qutrit (PennyLane-0.27.0)
- null.qubit (PennyLane-0.27.0)
- lightning.qubit (PennyLane-Lightning-0.27.0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.9.0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.9.0)
```

Existing GitHub issues

- [x] I have searched existing GitHub issues to make sure the issue does not already exist.

yitchen-tim added bug on Dec 7, 2022

soranjh on Dec 7, 2022 Contributor

Hi @yitchen-tim and thanks for the report. You get this error because the geometry array you created with `pennylane.numpy` has `requires_grad = True` by default. It means that the geometry is considered to be trainable and the initial values of the differentiable parameters should have the same shape of the geometry array. In your example, the geometry is differentiable but the initial values provided, `np.array(mol.alpha)`, are the basis function exponents.

The error disappears if you do one of the following options:

1. Set `requires_grad = False` for geometry and use `ei = qml.qchem.electron_integrals(mol)()`

```
import pennylane as qml
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symbols = ["H", "O", "H"]
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mol = qml.qchem.Molecule(symbols, geometry, basis_name="sto-3g")
ei = qml.qchem.electron_integrals(mol)()
```

2. Define correct initial values as `args = [geometry]` and use `ei = qml.qchem.electron_integrals(mol)(*args)` as

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ei = qml.qchem.electron_integrals(mol)(*args)
```

You can find a detailed description of how to define the initial input parameters and their meanings in this demo. The code examples given for each function, for example here, might be also helpful.
Please let me know if you have any other questions or if you get any other errors. Thanks!

soranjh on Dec 7, 2022 Contributor ···

Hi @yitchen-tim and thanks for the report. You get this error because the geometry array you created with `pennylane.numpy` has `requires_grad = True` by default. It means that the geometry is considered to be trainable and the initial values of the differentiable parameters should have the same shape of the geometry array. In your example, the geometry is differentiable but the initial values provided, `np.array(mol.alpha)`, are the basis function exponents.

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yitchen-tim on Dec 8, 2022 Author ···

Thanks for answering! I understand these functions now. In the examples of these functions, sometimes the input is geometry and sometime is alpha. I now understand that I put whatever I want to differentiate as input.

👍 1

lillian542 closed this as completed on Dec 12, 2022